WORKSHEET

Chapter 2: Collecting like terms

Simplify each of these algebraic expressions.

1 $x + 5x$

2 $3y + 8y$

3 $6a - a$

4 $9t - 4t$

5 $3ab + 6ab$

6 $8d + 12d$

7 $7st + 3st$

8 $2x + x$

9 $5a^2 + 4a^2$

10 $3t - 2t$

11 $4ab + 4ab$

12 $5ty - 3ty$

13 $2mp - mp$

14 $x + 4x - 3x$

15 $w + w + 5w$

16 $2c - 5c$

17 $3r - 3r$

18 $2m + 3m - m$

19 $e - 2e$

20 $14h + 6 - 12h$

21 $y + y + y$

22 $4c + d + 4c - 3d$

23 $7p + 2 + 3p + 7$

24 $5s - s + 2s$

25 $4y^2 + y^2$

26 $3d + d - d$

27 $a + a + 2f$

28 $c + 12d + 2c - 5d$

29 $6 + 3h + 4h + 1$

30 $6u^2 - 3u^2 + 2u^2$

31 $4a + 2a - 6b - 4b$

32 $5g - 4 + 2g + 3$

33 $10x + 6b - 5b - 7x$

34 $12m + 2n + 8m - 7n$

35 $-4a + 8a^2 + 5a + 3a^2$

36 $y + z + y - z$

37 $12w - 6w + 4wt + 3wt$

38 $4a + 6a + 2b - 3a$

39 $x^2 + 3y + y + x^2$

40 $6f + 3 + 2f + 1$

41 $5y^2 + 4x - 2y^2 + 3x$

42 $4k + p - 3k + p$

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43 $x + y - x$

44 $8y - 2 - 6y + 1$

45 $2y + 3m - y - m$

46 $-8 - 2q + 5q + 9$

47 $5r^2 + 3r^2 - r - 2r$

48 $3a - b + 3a$

49 $5t + 3t + 3v - 2v$

50 $18 - k + 4k - 10$

51 $14b - 2 + 3b - 6$

52 $4y + 2z - 3y + z$

53 $2x + 5 - 3x - 2$

54 $2ab + ab + 3b$

55 $2p^2 + 3p - p^2 + p$

56 $3u^2 + 5 - u^2 - 4$

57 $-5a + 2 + 3a + 3$

58 $8x - x - 8$

59 $10 + 6x + 3x - 7$

60 $9p - 3q - 7p - 2q$

61 $8t + 4w - 8t + 2w$

62 $15x + 3x - 5 + 5$

63 $-4p + 6p + a - 4a$

64 $6c + 7d - d - 5c$

65 $9jk - 5k - 2kj + 2k$

66 $5b - 5 - 12b + 5$

67 $-3x + 7y - 7y + 6x$

68 $-5y^2 + yz + 3yz - 2y^2$

69 $2y + 5 - 6y + 2$

70 $w^2 - 6u + 2w^2 + 9u$

71 $2bc + 4cd - bc + 3cd$

72 $-8ef + 2f^2 + 5f^2 + 10ef$

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Answers

1 $6x$
2 $11y$
3 $5a$
4 $5t$
5 $9ab$
6 $20d$
7 $10st$
8 $3x$
9 $9a^2$
10 t
11 $8ab$
12 $2ty$
13 mp
14 $2x$
15 $7w$
16 $-3c$
17 0
18 $4m$
19 $-e$
20 $2h + 6$
21 $3y$
22 $8c - 2d$
23 $10p + 9$
24 $6s$
25 $5y^2$
26 $3d$
27 $2a + 2f$
28 $3c + 7d$
29 $7h + 7$
30 $5u^2$
31 $6a - 10b$
32 $7g - 1$
33 $3x + b$
34 $20m - 5n$
35 $11a^2 + a$
36 $2y$
37 $6w + 7wt$
38 $7a + 2b$
39 $2x^2 + 4y$
40 $8f + 4$
41 $3y^2 + 7x$
42 $k + 2p$
43 y
44 $2y - 1$
45 $y + 2m$
46 $3q + 1$
47 $8r^2 - 3r$
48 $6a - b$
49 $8t + v$
50 $3k + 8$
51 $17b - 8$
52 $y + 3z$
53 $-x + 3$
54 $3ab + 3b$
55 $p^2 + 4p$
56 $2u^2 + 1$
57 $-2a + 5$
58 $7x - 8$
59 $9x + 3$
60 $2p - 5q$
61 $6w$
62 $18x$
63 $2p - 3a$
64 $c + 6d$
65 $7jk - 3k$
66 $-7b$
67 $3x$
68 $-7y^2 + 4yz$
69 $-4y + 7$
70 $3w^2 + 3u$
71 $bc + 7cd$
72 $7f^2 + 2ef$

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